Algebraic Number Theory NMAL431
Exercise Sheet 1 — Solutions

06 October 2026

Please send any comments and/or corrections to robin.visser@matfyz.cuni.cz

1. (a) Proven in the first lecture.

   (b) Clearly A_v is a subring of K, as it contains 1 and -1, and is closed under multiplication and addition by the valuation properties.

   Note that $A_v^\times = \{x \in K^\times : v(x) = 0\}$, as the element x and x^{-1} both lie in A_v if and only if $v(x) = 0$.

   By surjectivity, we can choose some uniformizer $\pi \in K^\times$ such that $v(\pi) = 1$. We note that every element $x \in A_v$ can be written in the form $x = u\pi^n$ with $n = v(x)$ and $v(u) = 0$. Thus every non zero ideal of A_v is of the form $\pi^n A_v$. Hence A_v is a PID with unique non-zero prime ideal πA_v.

   Finally, $\mathrm{Frac}A_v = K$, as for any $x \in K^\times$, either $v(x) \geq 0$ or $v(1/x) \geq 0$, and thus either $x \in A_v$ or $1/x \in A_v$.

   (c) We check that the two compositions are the identity.

   Let A be a DVR. Write $x \in K^\times$ as $x = u\pi^n$. If $n \geq 0$ then clearly $x \in A$. Conversely, if $x \in A$ and $n < 0$, then $\pi^{-1} = xu^{-1}\pi^{-n-1} \in A$, so π would be a unit, a contradiction. Hence $A = \{x \in K^\times : v_A(x) \geq 0\} \cup \{0\} = A_{v_A}$.

   Now let v be a valuation on K. By (b), any π with $v(\pi) = 1$ generates the maximal ideal of A_v, and $A_v^\times = \{v = 0\}$. Any $x \in K^\times$ can be written $x = u\pi^n$ with $u \in A_v^\times$, namely with $n = v(x)$ and $u = x\pi^{-n}$. Then $v_{A_v}(x) = n = v(x)$.

   Therefore $A \mapsto v_A$ and $v \mapsto A_v$ are mutually inverse bijections.

2. (a) Clearly v_p is well-defined by unique factorisation in $\mathbb{Z}$. To check that v_p is a homomorphism, let $x = p^n \frac{r}{s}$, and let $y = p^m \frac{r'}{s'}$, then:

$$v_p(xy) = v_p(p^n \frac{r}{s} \cdot p^m \frac{r'}{s'}) = v_p(p^{n+m} \frac{rr'}{ss'}) = n + m = v_p(x) + v_p(y)$$

   It's surjective as $v_p(p) = 1$. To check the ultrametric inequality, We note that

$$x + y = p^n \cdot \frac{rs' + p^{m-n} r's}{ss'}.$$

   The numerator is a non-zero integer, say $p^k t$ with $k \geq 0$ and $p \nmid t$, and $p \nmid ss'$. Hence $v_p(x + y) = n + k \geq n$.

(b) Let $v : \mathbb{Q}^\times \to \mathbb{Z}$ be a valuation. We want to show that $v = v_p$ for some prime p. Since we know v is a homomorphism, it suffices to just determine $v(q)$ for all primes q. Recall from lectures that we can define the sets $A \subset \mathbb{Q}$ and $P \subset \mathbb{Q}$ as

$$A = \{x \in \mathbb{Q}^\times \mid v(x) \geq 0\} \cup \{0\},$$
$$P = \{x \in \mathbb{Q}^\times \mid v(x) > 0\} \cup \{0\}.$$

From Question 1 (b), we know that A is a DVR and $P \subset A$ is the unique maximal ideal of A

As v is a homomorphism, we have that $v(1) = 0$ and $2v(-1) = v((-1)^2) = v(1) = 0$. This implies that $v(n) \geq v(\pm 1) = 0$ for all integers $n \in \mathbb{Z}$. Thus $\mathbb{Z} \subset A$.

As P is a prime ideal in A, thus $P \cap \mathbb{Z}$ is a prime ideal of $\mathbb{Z}$. If $P \cap \mathbb{Z} = \{0\}$, then we must have that $v(x) = 0$ for all nonzero integers $x \in \mathbb{Z}$. But this implies that $v(r/s) = v(r) - v(s) = 0$ for any $r, s \in \mathbb{Z}$, $r, s \neq 0$. This implies v is the zero map, which contradicts that v is surjective.

Therefore P is a nonzero prime ideal of $\mathbb{Z}$, and so $P \cap \mathbb{Z} = (p)$ for some prime number p. For any prime $q \neq p$, we have $q \notin P \cap \mathbb{Z}$ which implies $v(q) = 0$.

For any positive integer $n \in \mathbb{N}$, let $n = \prod_{i=1}^k p_i^{a_i}$. Then we have that

$$v(n) = \sum_{i=1}^k a_i v(p_i) = \begin{cases} 0 & \text{if } p \nmid n, \\ v_p(n)v(p) & \text{if } p \mid n, \end{cases}$$

where we've used that $a_i = v_{p_i}(n)$ by definition. As v is surjective, we must have $v(p) = 1$, and thus $v = v_p$. □

3. (a) Suppose $0 \in S$. Then for any $a/s \in S^{-1}A$, we have tat $a/s \sim 0/1$ as $0 \cdot (a \cdot 1 - 0 \cdot s) - 0$. Hence $S^{-1}A = 0$.

Conversely, suppose $S^{-1}A = 0$, then $1/1 = 0/1$. Then there exists some $t \in S$ such that $t(1 \cdot 1 - 0 \cdot 1) = 0$. Thus $t = 0$, so $0 \in S$.

(b) Let $R \subset \mathbb{Q}$ be a subring. We have that $\mathbb{Z} \subset R$ as $1 \in R$. Define the set

$$S := \{s \in \mathbb{Z} \backslash \{0\} : 1/s \in R\}$$

Note that S is a multiplicatively closed set, as $1 \in S$, and if $s, t \in S$ then $st \in S$. We claim that $R = S^{-1}\mathbb{Z}$.

By Lemma 1.7, we can identify $S^{-1}\mathbb{Z}$ with the subring $\{a/s : a \in \mathbb{Z}, s \in S\}$ of $\mathbb{Q}$, which is clearly contained in R. For the reverse inclusion, let $x \in R$ and write $x = a/b$ in lowest terms. Since $\gcd(a, b) = 1$, there are $m, n \in \mathbb{Z}$ with $ma + nb = 1$. Dividing by b gives $1/b = mx + n \in R$. So $b \in S$ and $x \in S^{-1}\mathbb{Z}$. Hence $R = S^{-1}\mathbb{Z}$.

(c) This was proven in lectures; we'll recall the proof again here. Let $K = \mathrm{Frac}A$. Each $A_\mathfrak{m}$ is a subring of K containing A, so $A \subset \bigcap_\mathfrak{m} A_\mathfrak{m}$.

Now let $a \in \bigcap_\mathfrak{m} A_\mathfrak{m}$. This implies that, for all maximal ideals $\mathfrak{m}$, we have $a = x_\mathfrak{m}/s_\mathfrak{m}$ for some $x_\mathfrak{m} \in A$, $s_\mathfrak{m} \in A - \mathfrak{m}$. Now consider the ideal

$I = (s_\mathfrak{m} \mid \mathfrak{m}$ maximal ideal). This cannot be contained in any maximal ideal, and thus $I = A$. Therefore we have $1 = \sum_\mathfrak{m} t_\mathfrak{m} s_\mathfrak{m}$ for some $t_\mathfrak{m} \in A$ which are 0 for all but finitely many $\mathfrak{m}$. Thus $a = \sum_\mathfrak{m} t_\mathfrak{m} s_\mathfrak{m} a \in A$.

(If A is a field, its only maximal ideal is 0 and $A_0 = K = A$, so the statement is trivial; the argument above covers this case anyway.)

(d) On the left-hand side, $\bigcap_n 2^n\mathbb{Z} = 0$, because a non-zero integer is divisible by only finitely many powers of 2. So $S^{-1}\big(\bigcap_n 2^n\mathbb{Z}\big) = S^{-1}(0) = 0$.

On the right-hand side, $S^{-1}\mathbb{Z} = \mathbb{Z}[1/2]$. By exactness of localisation (Lemma 1.9), $S^{-1}(2^n\mathbb{Z})$ is the ideal $2^n\mathbb{Z}[1/2]$ of $\mathbb{Z}[1/2]$. This is all of $\mathbb{Z}[1/2]$, because 2 is a unit there. So

$$\bigcap_n S^{-1}(2^n\mathbb{Z}) = \mathbb{Z}[1/2] \neq 0.$$

The moral is that localisation commutes with finite intersections (as used in Lemma 1.13), but not with infinite ones.

4. See Atiyah–Macdonald's textbook *Introduction to Commutative Algebra*, Proposition 1.10.

5. (a) A PID is always a UFD; this is standard and does not need the Dedekind hypothesis. E.g. See Lang's *Algebra* textbook, II.5.2.

Conversely, suppose that R is a UFD. As every ideal factors uniquely as a product of prime ideals, it suffices to show that every non-zero prime $\mathfrak{p}$ is principal. Pick a non-zero $x \in \mathfrak{p}$. Since $\mathfrak{p}$ is proper, x is not a unit, so it factors as $x = uq_1 \cdots q_k$ with $u \in R^\times$ and each q_i a prime element. Since $\mathfrak{p}$ is prime, some $q_i \in \mathfrak{p}$.

Now (q_i) is a non-zero prime ideal with $(q_i) \subset \mathfrak{p}$. Non-zero primes of a Dedekind domain are maximal, so $(q_i) = \mathfrak{p}$. Hence $\mathfrak{p}$ is principal.

(b) Let R be a Dedekind domain, and let all the nonzero maximal ideals of R be $P_1, P_2, \ldots, P_n$. We first show that P_1 is principal. We can use the Chinese remainder theorem on the set of ideals $P_1^2, P_2, \ldots, P_r$. We note these are pairwise coprime. We choose some $\alpha \in R$ such that

- $\alpha \in P_1 - P_1^2$
- $\alpha \notin P_i$ for all $i = 2, \ldots, r$.

As R is a Dedekind domain, we know from lectures that (α) has a unique prime factorisation into ideals as $(\alpha) = \prod_i P_i^{v_{P_i}(\alpha)}$, which implies $(\alpha) = P_1$, and thus P_1 is principal.

By analogous arguments for every other prime ideal $P_2, \ldots, P_r$, it follows that all prime ideals are principal, and thus by factoring, it follows that all ideals are principal. □

(c) Let R be a Dedekind domain and let $I \subset R$ be any nonzero ideal. Pick some $\alpha \in I - \{0\}$. As $(\alpha) \subset I$, by factoring the ideal I, we have that there exists some ideal $J \subset R$ such that $(\alpha) = IJ$. We now again use the Chinese remainder theorem to pick some $\beta \in R$ such that

- $v_P(\beta) = v_P(I)$ for all prime ideals P dividing I.

- $v_P(\beta) = 0$ for all prime ideals P dividing (α) but not I.

We claim that $I = (\alpha, \beta)$. This will follow by showing that $v_P((\alpha, \beta)) = v_P(I)$ for all primes $P \subset R$, and applying unique factorisation.

To prove the claim, we first note that $(\alpha, \beta) = (\alpha) + (\beta)$ and thus that $v_P((\alpha, \beta)) \geq \min(v_P(\alpha), v_P(\beta))$. However, by construction we have that $v_P(a) \neq v_P(\beta)$, and thus we have the equality $v_P((\alpha, \beta)) = \min(v_P(\alpha), v_P(\beta)) = v_P(I)$, which proves the claim. □

6. Let A be a Dedekind domain, $K = \mathrm{Frac}A$ and $f(X) \in A[X]$ a monic polynomial, $g_(X), g_2(X) \in K[X]$, For each prime ideal P of A, we have that A_P is a DVR, and thus principal. Then by Gauss's lemma, we have that $g_1(X), g_2(X) \in A_P[X]$, for every prime ideal P.

   As A is a Dedekind domain, every non-zero prime ideal is maximal, and thus by Q3 (c), we have that $\cap_P A_P = A$. This implies that $g_1(X), g_2(X) \in A[X]$. □